

Project Presentation

Information Technology Department



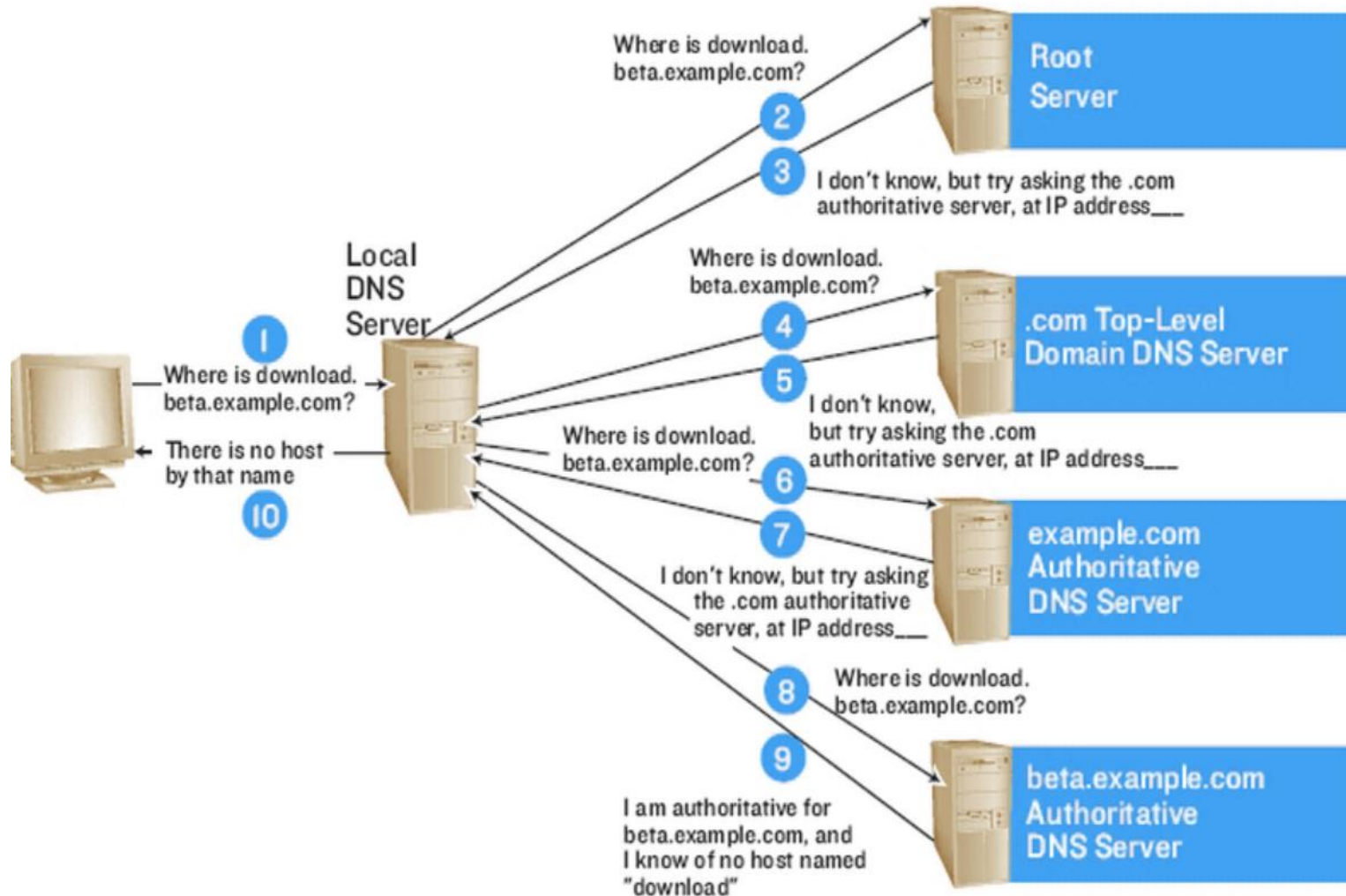
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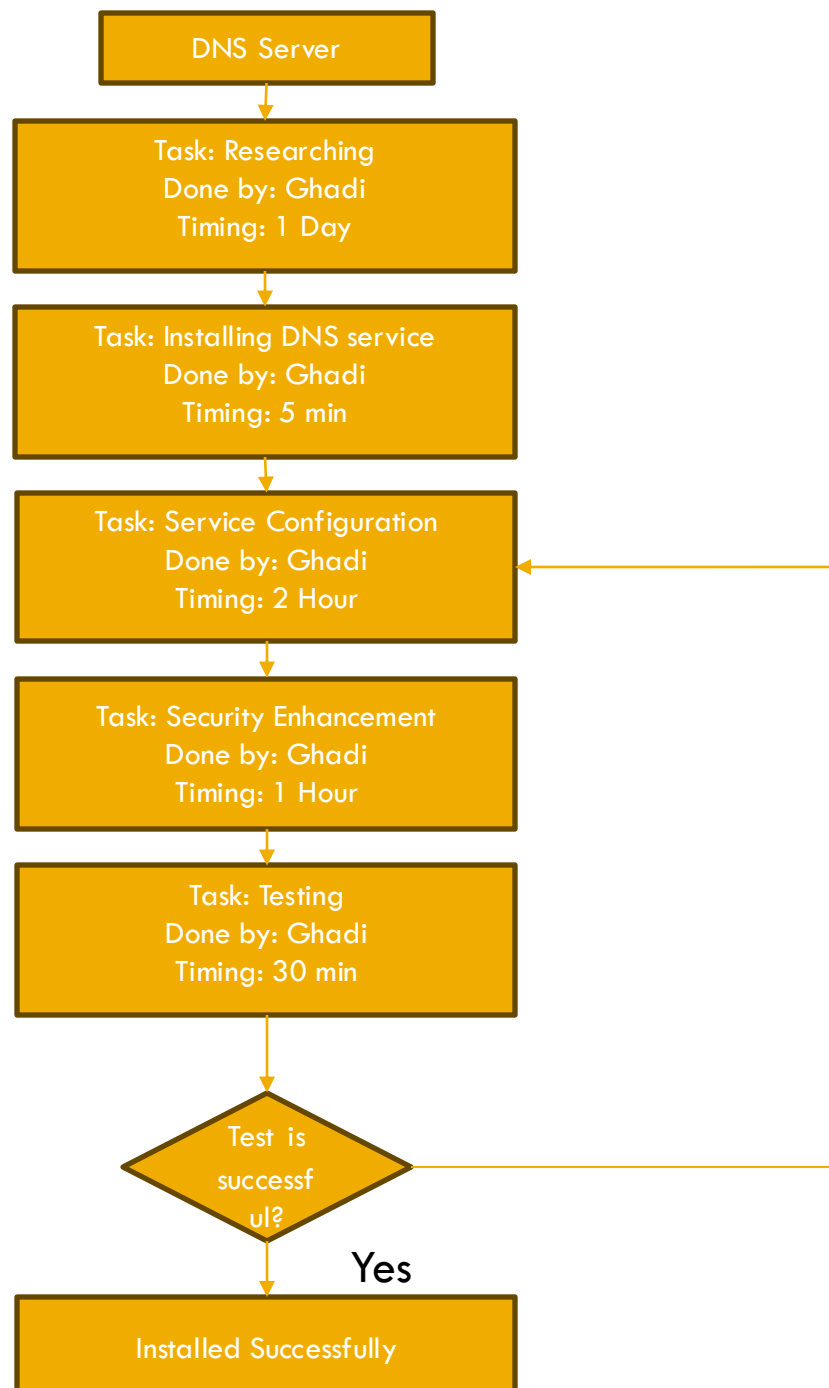
DNS Server



HOW DNS WORKS



2. Project organisation



3. Project realisation

DNS Implementation Overview:

Service Installation: Deploying the named (BIND9) service and its utilities on the Ubuntu Server environment.

Network Configuration: Setting up Forward and Reverse lookup zones to manage domain name resolution and IP mapping.

Security Enhancement: Hardening the server using Access Control Lists (ACLs) to restrict queries and enabling DNSSEC for data integrity.

3. Project realisation

1- Available and selected solution

List the available solutions:

- BIND9 (named): The most widely used DNS software on Linux.
- Unbound: A validating, recursive, caching DNS resolver.
- dnsmasq: A lightweight DNS forwarder and DHCP server.

Selected Solution: BIND9 (named).

Advantages of the selected solution:

- Full DNSSEC Support: Provides robust security and data integrity.
- Highly Scalable: Can handle complex zone configurations (Forward and Reverse).
- Standardized: It is the industry standard for Linux-based DNS infrastructure.

2- Install requirements and dependencies

OS Environment: Ubuntu Server (Virtual Machine).

Installed Packages:

- bind9: The main DNS server package.
- bind9utils: Utilities for testing and managing the DNS server.
- bind9-doc: Documentation for BIND.

Version: BIND9 (Latest Stable Version)

Installation Instruction: `sudo apt update & sudo apt install bind9 bind9utils bind9-doc -y`

1.Update package Repository

```
ghadi@ghadi-VirtualBox:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Hit:2 http://sa.archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://sa.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [14.6 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [7,692 B]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [792 B]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [704 B]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [7,472 B]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [4,764 B]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [332 B]
Hit:11 http://sa.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:12 http://sa.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [50.2 kB]
Get:13 http://sa.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [13.3 kB]
Get:14 http://sa.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [792 B]
Get:15 http://sa.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [872 B]
Get:16 http://sa.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [29.4 kB]
Get:17 http://sa.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [6,588 B]
Get:18 http://sa.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [7,472 B]
Get:19 http://sa.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [4,764 B]
Get:20 http://sa.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [332 B]
Fetched 421 kB in 2s (233 kB/s)
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
ghadi@ghadi-VirtualBox:~$
```

The purpose of this command is to update the list of packages available in the Ubuntu operating system from the official package repositories. This helps the system to know if there are new updates or updated versions of any package.

2. Install Bind

```
sghadi@ghadi-VirtualBox:~$ sudo apt install bind9 bind9utils bind9-doc -y
Note, selecting 'bind9-utils' instead of 'bind9utils'
Installing:
  bind9  bind9-doc  bind9-utils

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 9
  Download size: 3,701 kB
  Space needed: 9,449 kB / 13.9 GB available

Get:1 http://sa.archive.ubuntu.com/ubuntu resolute/main amd64 bind9-utils amd64 1:9.20.18-1ubuntu2 [185 kB]
Get:2 http://sa.archive.ubuntu.com/ubuntu resolute/main amd64 bind9 amd64 1:9.20.18-1ubuntu2 [259 kB]
Get:3 http://sa.archive.ubuntu.com/ubuntu resolute/main amd64 bind9-doc all 1:9.20.18-1ubuntu2 [3,257 kB]
Fetched 3,701 kB in 1s (2,934 kB/s)
Selecting previously unselected package bind9-utils.
(Reading database... 153812 files and directories currently installed.)
Preparing to unpack .../bind9-utils_1%3a9.20.18-1ubuntu2_amd64.deb...
Unpacking bind9-utils (1:9.20.18-1ubuntu2)...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_1%3a9.20.18-1ubuntu2_amd64.deb...
Unpacking bind9 (1:9.20.18-1ubuntu2)...
Selecting previously unselected package bind9-doc.
Preparing to unpack .../bind9-doc_1%3a9.20.18-1ubuntu2_all.deb...
Unpacking bind9-doc (1:9.20.18-1ubuntu2)...
Setting up bind9-doc (1:9.20.18-1ubuntu2)...
Setting up bind9-utils (1:9.20.18-1ubuntu2)...
Setting up bind9 (1:9.20.18-1ubuntu2)...
wrote key file "/etc/bind/rndc.key"
named-resolvconf.service is a disabled or a static unit, not starting it.
Created symlink '/etc/systemd/system/bind9.service' → '/usr/lib/systemd/system/named.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/named.service' → '/usr/lib/systemd/system/named.service'.
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for ufw (0.36.2-9build1)...
sghadi@ghadi-VirtualBox:~$
```

This command downloads and installs the bind9 package, which is required to run the DNS service on the system. and The image shows that the bind9 package is **already installed** with the latest version and does not need to be reinstalled

3. Project realisation

3.Enable Service persistence

```
ghadi@ghadi-VirtualBox:~$ sudo systemctl enable named
Synchronizing state of named.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable named
Created symlink '/etc/systemd/system/bind9.service' -> '/usr/lib/systemd/system/named.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/named.service' -> '/usr/lib/systemd/system/named.service'.
ghadi@ghadi-VirtualBox:~$ sudo systemctl start named
```

Ensuring service continuity by creating the systemd symlinks. This step guarantees that the DNS infrastructure is persistent and reliable across server reboots

Step 1: Update System Repositories (sudo apt update).

Step 2: Install DNS Suite (sudo apt install bind9 bind9utils)

Step 3: Enable Service Persistence (sudo systemctl enable named).

Step 4: Start & Verify Status (sudo systemctl status named)

4.Reverse lookup zone

```
ghadi@ghadi-VirtualBox: ~ — sudo nano /etc/bind/db.reverse
GNU nano 8.7.1 /etc/bind/db.reverse
$TTL      604800
@         IN      SOA      mydomain.local. root.mydomain.local. (
                                2
                                604800
                                86400
                                2419200
                                604800 )
@         IN      NS       ns.mydomain.local.
15        IN      PTR      mydomain.local.
15        IN      PTR      ns.mydomain.local.
```

We configured the reverse lookup zone to map IP addresses to domain names using PTR records

5. Forward lookup zone

```
ghadi@ghadi-VirtualBox: ~ — sudo nano /etc/bind/db.mydomain
GNU nano 8.7.1 /etc/bind/db.mydomain
$TTL      604800
@         IN      SOA      mydomain.local. root.mydomain.local. (
                        2      ; Serial
                        604800 ; Refresh
                        86400  ; Retry
                        2419200 ; Expire
                        604800 ) ; Negative Cache TTL
@         IN      NS       ns.mydomain.local.
ns        IN      A        10.0.2.15
www       IN      A        10.0.2.15
```

We configured the forward lookup zone to map domain names to IP addresses using a records

6. Validate the syntax

```
ghadi@ghadi-VirtualBox:~$ sudo nano /etc/bind/db.reverse
ghadi@ghadi-VirtualBox:~$ sudo nano /etc/bind/db.reverse
ghadi@ghadi-VirtualBox:~$ sudo named-checkzone 10.0.2.in-addr.arpa /etc/bind/db.reverse
[sudo: authenticate] Password:
zone 10.0.2.in-addr.arpa/IN: loaded serial 2
OK
```

Using the named-checkzone command to validate the syntax and integrity of the reverse lookup zone file (db.reverse)."

"The output **OK** confirms that the configuration is correct, free of syntax errors, and the zone is ready to be loaded by the BIND services

```
ghadi@ghadi-VirtualBox:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-30 21:21:53 +03; 8min ago
 Invocation: 9774929138814c96ae0f3d65867a7adf
    Docs: man:named(8)
  Main PID: 6821 (named)
   Status: "running"
    Tasks: 11 (limit: 1841)
  Memory: 6.8M (peak: 7.2M)
     CPU: 48ms
   CGroup: /system.slice/named.service
           └─6821 /usr/sbin/named -f -u bind

Apr 30 21:21:53 ghadi-VirtualBox named[6821]: command channel listening on 127.0.0.1#953
Apr 30 21:21:53 ghadi-VirtualBox named[6821]: configuring command channel from '/etc/bind/rndc.key'
Apr 30 21:21:53 ghadi-VirtualBox named[6821]: command channel listening on ::1#953
Apr 30 21:21:53 ghadi-VirtualBox named[6821]: managed-keys-zone: loaded serial 2
Apr 30 21:21:53 ghadi-VirtualBox named[6821]: all zones loaded
Apr 30 21:21:53 ghadi-VirtualBox named[6821]: FIPS mode is disabled
Apr 30 21:21:53 ghadi-VirtualBox named[6821]: running
Apr 30 21:21:53 ghadi-VirtualBox systemd[1]: Started named.service - BIND Domain Name Server.
Apr 30 21:21:53 ghadi-VirtualBox named[6821]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance timer complete)
Apr 30 21:21:53 ghadi-VirtualBox named[6821]: managed-keys-zone: Key 38696 for zone . is now trusted (acceptance timer complete)
ghadi@ghadi-VirtualBox:~$
```

The final verification output showing the **named** service in an **active (running)** state. All zones are confirmed loaded, and the system is ready to resolve domain names

3. Project realisation

7- Service configuration

The main configuration file for BIND is `/etc/bind/named.conf`. Additional configurations can be placed in the files `/etc/bind/named.conf.options` and `/etc/bind/named.conf.local` to further customize and fine-tune the server according to the specific environment needs

DNS Forwarding:

```
ghadi@ghadi-VirtualBox: ~ — sudo nano /etc/bind/named.conf.options
GNU nano 8.7.1 /etc/bind/named.conf.options
options {
    directory "/var/cache/bind";
    recursion yes;

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };
};
```

Configuring Forwarders to enable the server to resolve external domain names

ghadi@ghadi-VirtualBox: ~ — sudo nano /etc/bind/named.conf.local

```
GNU nano 8.7.1 /etc/bind/named.conf.local
zone "mydomain.local" {
    type master;
    file "/etc/bind/db.mydomain";
};

zone "2.0.10.in-addr.arpa" {
    type master;
    file "/etc/bind/db.reverse";
};
//
// Do any local configuration here
//
```

Defining the authoritative Forward and Reverse zones for our local domain

3. Project realisation

8.Security enhancement

```
ghadi@ghadi-VirtualBox: ~ -- sudo nano /etc/bind/named.conf.options
GNU nano 8.7.1 /etc/bind/named.conf.options
acl "trusted" {
    10.0.2.0/24;
};
options {
    directory "/var/cache/bind";
    recursion yes;
    allow-recursion { trusted; };
};
```

Implementing Access Control Lists (ACLs): Restricting DNS queries to trusted internal networks only to prevent unauthorized external access.

Recursive Query Limitation: Ensuring the server only performs recursion for authorized clients, which mitigates the risk of DNS amplification attacks.

Service Isolation: Running the **named** service within a dedicated virtual environment to isolate potential threats from the host system.

Access Control List Configuration. The screenshot illustrates the definition of a 'trusted' ACL that includes the local subnet (10.0.2.0/24). By applying the allow-recursion { trusted; }; directive, we ensure that the DNS server only responds to legitimate internal requests, significantly enhancing the system's security posture.

3. Project realisation

8- Security enhancement

there are many ways to enhance the security of the DNS server some of them are:

2- Configure Firewall

3- Use a non-open source resolver

4- Increase resilience with a private DNS network A private network offers three key benefits:

- Lower latency - Enhanced security
- Better reliability

5- Deploy DNS security extensions (DNSSEC) By implementing security measures, the security posture of the BIND DNS server can be improved and the risk of potential security threats and attacks reduced.

3. Project realisation

9.Test

Test Case: Reverse DNS Lookup Verification.

Objective: To ensure the DNS server correctly translates IP addresses into their associated hostnames.

Method: Executing the **nslookup** command for the local IP **10.0.2.15** using the local server address **127.0.0.1** as the DNS resolver.

Expected Result: The system should return the FQDN (Fully Qualified Domain Name) defined in the reverse zone file.

```
ghadi@ghadi-VirtualBox:~$ nslookup www.mydomain.local 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   www.mydomain.local
Address: 10.0.2.15

ghadi@ghadi-VirtualBox:~$
```

To verify that the DNS server can resolve a hostname to its correct IP address.

Result: The server **successfully mapped** www.mydomain.local to 10.0.2.15

3. Project realisation

9.Test

```
ghadi@ghadi-VirtualBox:~$ sudo nano /etc/bind/named.conf.local
ghadi@ghadi-VirtualBox:~$ sudo systemctl restart named
ghadi@ghadi-VirtualBox:~$ nslookup 10.0.2.15 127.0.0.1
15.2.0.10.in-addr.arpa  name = ns.mydomain.local.
15.2.0.10.in-addr.arpa  name = mydomain.local.

ghadi@ghadi-VirtualBox:~$
```

To verify the reverse mapping functionality.

Result: The server correctly identified the hostname ns.mydomain.local for the IP 10.0.2.15.

4. Conclusion:

In this project, we successfully installed and configured a DNS server using BIND on Ubuntu.

We implemented forward and reverse lookup zones to enable domain name resolution.

Security measures such as ACL and DNSSEC were applied to protect the server.

We also evaluated performance through DNS caching and tested the system using nslookup.

Overall, the DNS server worked efficiently and met all project requirements.

5. References

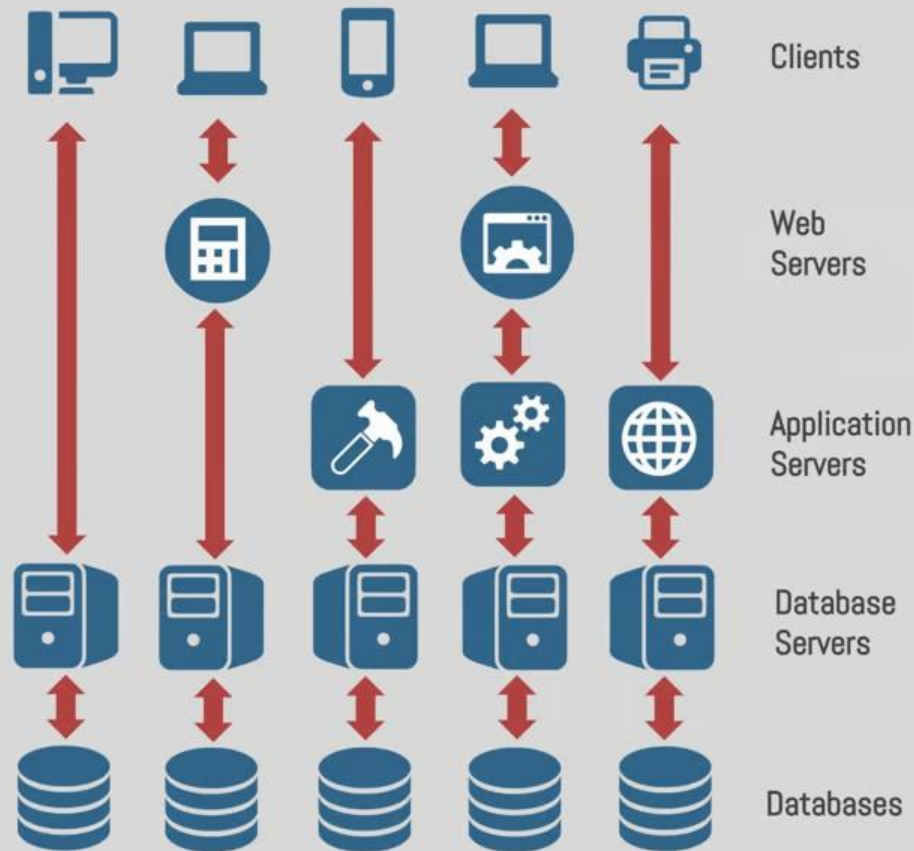
- [Dns Server And Web Server Simulation With Debian Operating System On Local Area Network | Jurnal Media Computer Science](#)
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Database Server

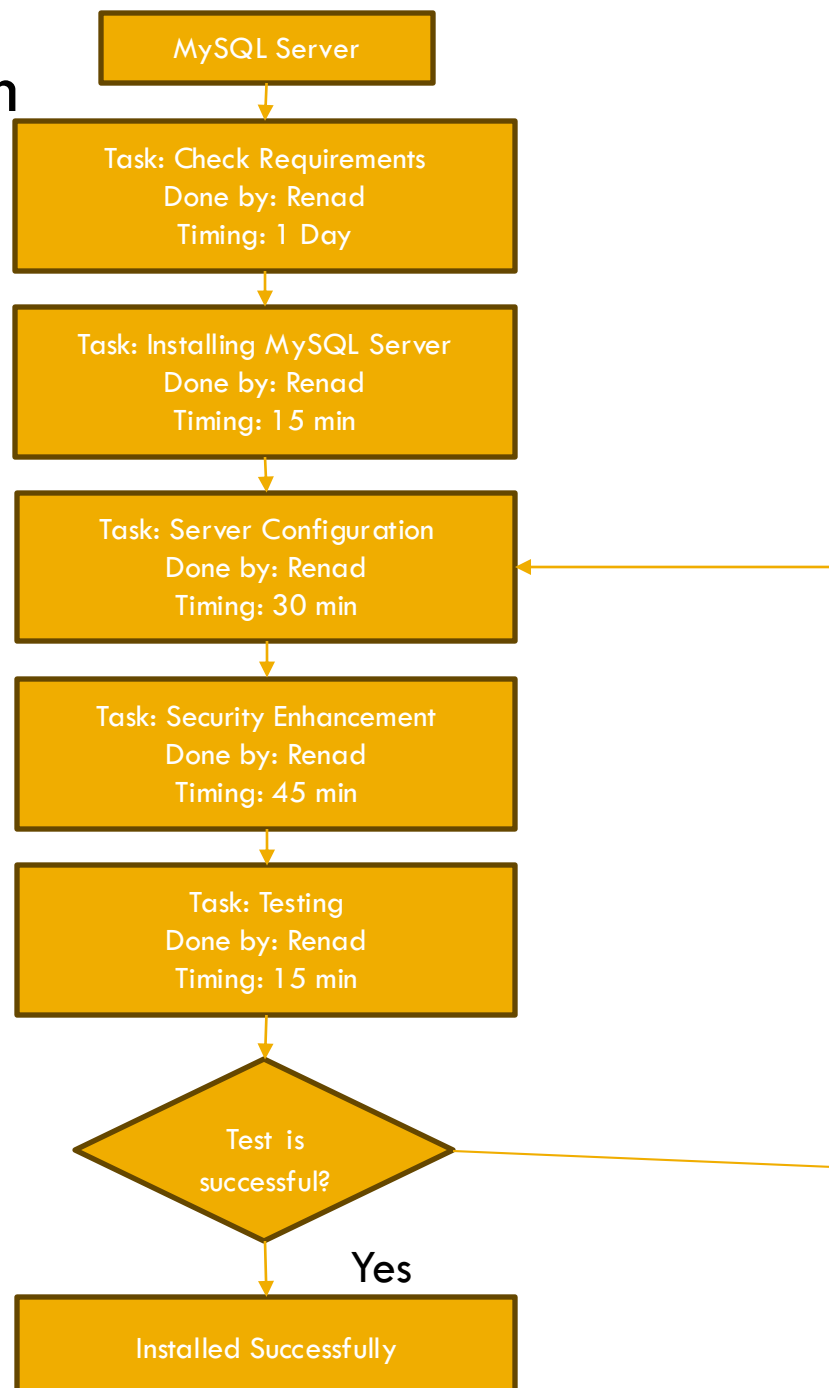


1. How database server work

Database Servers in Action



2. Project organisation



2. Project organisation

A database server is a system that runs database management software and is responsible for storing, managing, and retrieving data in an organized and efficient way. It plays a key role in modern applications, where client systems send requests to the server to access or update data.

In a client-server architecture, the database server acts as a central component that ensures data consistency, availability, and controlled access. It supports multiple users and applications at the same time while maintaining performance and reliability.

In this project, a database server is implemented in a Linux environment using MySQL. The project focuses on demonstrating the essential steps required to deploy the service, including installation, configuration, basic security measures, and testing. The goal is to show how a database server can be set up and used effectively in a real system environment.

2. Project organisation

There are several database management systems available for deployment on Linux systems, each with different features and use cases. Common examples include:

PostgreSQL, known for its advanced features and strong support for complex queries MongoDB, which is a NoSQL database designed for flexible and unstructured data MySQL, one of the most widely used relational database systems

For this project, MySQL was selected as the database solution. It is an open-source database management system that is widely used in web applications and enterprise environments. MySQL is known for its simplicity, reliability, good performance, and strong community support, making it suitable for beginners and practical implementations.

To set up the database server, the MySQL package is installed using Linux package management tools through the terminal. This process automatically installs all required dependencies, libraries, and components needed for the database server to function correctly. After installation, the system becomes ready for further configuration, security setup, and testing.

1. Install MySQL:

Step 1: Update System Packages

```
renad@ubuntu24:~$ sudo apt update
[sudo] password for renad:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://sa.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://sa.archive.ubuntu.com/ubuntu noble-updates InRelease
Get:4 http://sa.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://sa.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [64.5 kB]
Get:6 http://sa.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:7 http://sa.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [34.1 kB]
Get:8 http://sa.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1,484 B]
Fetched 227 kB in 2s (104 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
182 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Refreshes the system's package list from the repositories. This ensures that the latest available versions of software and security updates are recognized before installing any new package.

1. Install MySQL:

Step 2: Install MySQL Server

```
renad@ubuntu24:~$ sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

Installs the MySQL server on the system, including all required components and dependencies. Once completed, the system will have a fully functional database service ready for configuration.

Step 3: Verify Installation

```
renad@ubuntu24:~$ mysql --version
mysql Ver 8.0.45-0ubuntu0.24.04.1 for Linux on x86_64 ((Ubuntu))
```

Checks that MySQL has been installed correctly by displaying the installed version. This confirms that the installation process was successful.

2. Service Configuration

Access MySQL

```
renad@ubuntu24:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```

Opens the MySQL command-line interface using administrative privileges, allowing full control over database management and configuration.

2. Service Configuration

1-Create Database

```
mysql> CREATE DATABASE project_db;  
Query OK, 1 row affected (0.11 sec)
```

This step creates a new database that will be used to store application data. The database acts as a container for tables, records, and all related data. It is an essential step before performing any operations such as inserting or retrieving data.

2-Create User

```
mysql> CREATE USER 'user1'@'localhost' IDENTIFIED BY 'user123';  
Query OK, 0 rows affected (0.05 sec)
```

This command creates a dedicated user account for accessing the database. Instead of using the root account for all operations, creating a separate user improves security and allows better control over permissions.

2. Service Configuration

Grant Privileges

```
mysql> GRANT ALL PRIVILEGES ON project_db.* TO 'user1'@'localhost';  
Query OK, 0 rows affected (0.10 sec)  
  
mysql> FLUSH PRIVILEGES;  
Query OK, 0 rows affected (0.01 sec)
```

This step grants the created user full access to the database. It allows the user to perform operations such as creating tables, inserting data, updating records, and deleting entries. The FLUSH PRIVILEGES command ensures that all changes are applied immediately.

3. Security Enhancemen

Secure MySQL Installatio

```
renad@ubuntu24:~$ sudo mysql_secure_installation
[sudo] password for renad:

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.
The 'validate_password' component is installed on the server.
The subsequent steps will run with the existing configuration
of the component.

Skipping password set for root as authentication with auth_socket is used by default.
If you would like to use password authentication instead, this can be done with the "ALTER_USER" command.
See https://dev.mysql.com/doc/refman/8.0/en/alter-user.html#alter-user-password-management for more information.

By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
environment.

Remove anonymous users? (Press y|Y for Yes, any other key for No) : y
Success.

Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : y
Success.

By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.
```

```
Remove test database and access to it? (Press y|Y for Yes, any other key for No) : y
- Dropping test database...
Success.

- Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y
Success.

All done!
renad@ubuntu24:~$
```

Runs a built-in security script that enhances the protection of the MySQL server. It removes anonymous users, restricts root access, and applies basic security settings to prevent unauthorized access. **This step is sufficient to meet the required database security.**

4. Backup

Backup Database

```
renad@ubuntu24:~$ sudo mysqldump -u root -p project_db > backup.sql  
Enter password:  
renad@ubuntu24:~$
```

Creates a backup file of the database. This ensures that data can be restored in case of system failure, accidental deletion, or data corruption.

5. Testing

Check MySQL Service Status

```
renad@ubuntu24:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-05-02 11:24:50 UTC; 55min ago
     Process: 1267 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 1320 (mysqld)
      Status: "Server is operational"
        Tasks: 39 (limit: 2476)
       Memory: 374.6M (peak: 479.2M)
          CPU: 1min 1.894s
         CGroup: /system.slice/mysql.service
                └─1320 /usr/sbin/mysqld

May 02 11:24:48 ubuntu24 systemd[1]: Starting mysql.service - MySQL Community Server...
May 02 11:24:50 ubuntu24 systemd[1]: Started mysql.service - MySQL Community Server.
```

Checks whether the MySQL service is running correctly. It confirms that the database server is active and ready to handle requests.

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| project_db |
| sys |
+-----+
5 rows in set (0.03 sec)
```

Displays all databases to confirm that the created database exists.

References:

<https://www.thegeekstuff.com/2014/01/install-dns-server/>

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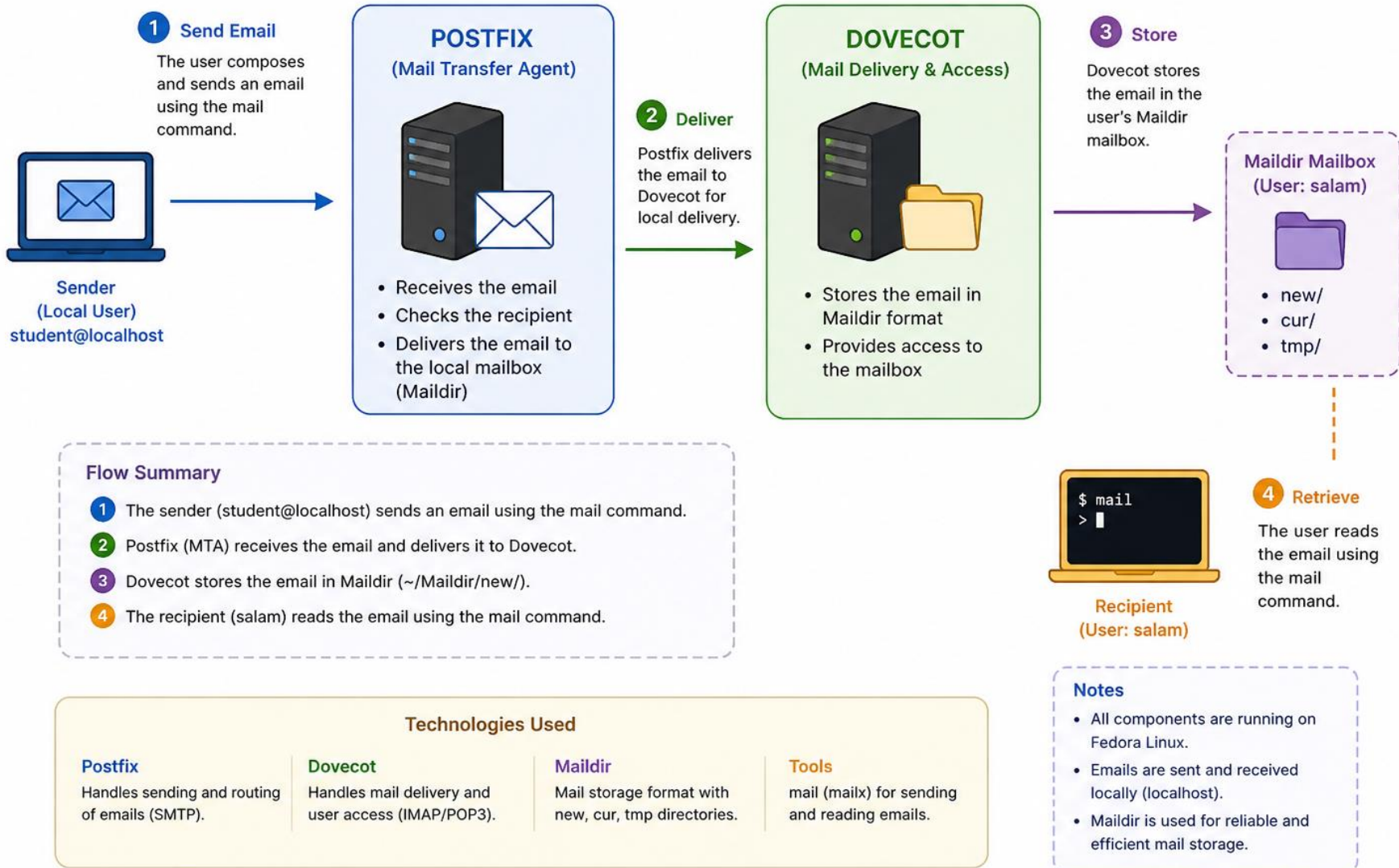
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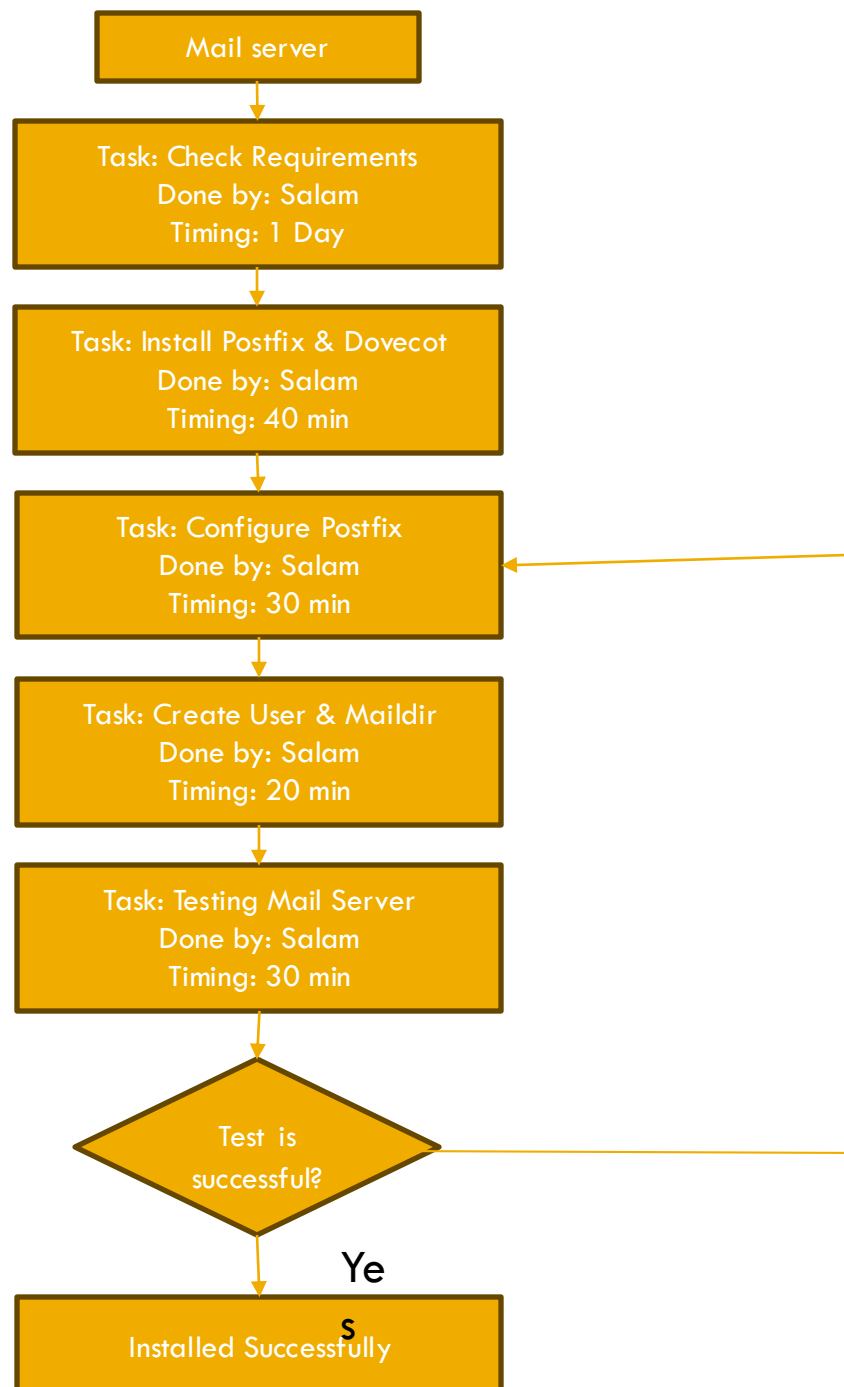
Mail Server



1. How Mail server work



2. Project organisation



3.Project Realization – Mail Server

The mail server was successfully implemented using Fedora Linux. The project began by installing the required packages, including Postfix, Dovecot, and Mailx. Postfix was configured as the mail transfer agent to handle sending emails, while Dovecot was configured to manage mail delivery and user access.

A local user account was created, and the Maildir structure was set up to store emails efficiently. The system was configured to ensure proper communication between Postfix and Dovecot.

After completing the configuration, several tests were conducted by sending emails locally. The results confirmed that emails were successfully delivered and stored in the user's mailbox.

Overall, the project achieved its objective of building a functional and reliable mail server capable of sending, receiving, and storing emails.

3.Project Realization – Mail Server

•System Update

- The system was updated to ensure all packages are up to date before installation.

```
student@localhost:~  
File Edit View Search Terminal Help  
[student@localhost ~]$ sudo dnf update -y  
[sudo] password for student:  
Sorry, try again.  
[sudo] password for student:  
Last metadata expiration check: 0:06:45 ago on Tue 28 Apr 2026 10:36:15 AM +03.  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repository	Size
Upgrading:				
LibRaw	x86_64	0.19.5-1.fc29	updates	303 k
NetworkManager	x86_64	1:1.12.6-5.fc29	updates	1.6 M
NetworkManager-adsl	x86_64	1:1.12.6-5.fc29	updates	26 k
NetworkManager-bluetooth	x86_64	1:1.12.6-5.fc29	updates	46 k
NetworkManager-config-connectivity-fedora	noarch	1:1.12.6-5.fc29	updates	14 k
NetworkManager-libnm	x86_64	1:1.12.6-5.fc29	updates	1.2 M
NetworkManager-openconnect	x86_64	1.2.6-2.fc29	updates	454 k
NetworkManager-openconnect-gnome	x86_64	1.2.6-2.fc29	updates	41 k
NetworkManager-openvpn	x86_64	1:1.8.10-1.fc29	updates	247 k
NetworkManager-openvpn-gnome	x86_64	1:1.8.10-1.fc29	updates	52 k

```
...  
) : java-openjdk-headless-10.0.2.13-7.fc 20 kB/s | 40 MB 34:18  
-----  
645 kB/s | 1.3 GB 35:11  
...  
transaction check  
ion check succeeded.  
transaction test  
ion test succeeded.  
transaction  
g scriptlet: java-openjdk-headless-1:10.0.2.13-7.fc29.x86_64 1/1  
g scriptlet: firefox-70.0-1.fc29.x86_64 1/1  
g scriptlet: java-11-openjdk-headless-1:11.0.5.10-0.fc29.x86_64 1/1  
ing : 1/1  
g scriptlet: libgcc-8.3.1-2.fc29.x86_64 1/1  
libgcc-8.3.1-2.fc29.x86_64  
ing : libgcc-8.3.1-2.fc29.x86_64 1/1664  
g scriptlet: libgcc-8.3.1-2.fc29.x86_64 1/1664  
libgcc-8.3.1-2.fc29.x86_64  
ed: linux-firmware-whence-20191022-103.fc29.noarch  
ling : linux-firmware-whence-20191022-103.fc29.noarch 2/1664  
ed: linux-firmware-whence-20191022-103.fc29.noarch  
: tzdata-2019c-1.fc29.noarch  
ding : tzdata-2019c-1.fc29.noarch 3/1664  
: tzdata-2019c-1.fc29.noarch  
: pcre2-10.33-15.fc29.x86_64  
ding : pcre2-10.33-15.fc29.x86_64 4/1664
```

3.Project Realization – Mail Server

1.Installation

The required packages (Postfix, Dovecot, and Mailx) were installed to set up the mail

```
python3-requests-2.31.0-1.fc29.x86_64.rpm
Complete!
[student@localhost ~]$ sudo dnf install postfix dovecot -y
[sudo] password for student:
Last metadata expiration check: 1:16:32 ago on Tue 28 Apr 2026 10:36:15 AM +03.
Dependencies resolved.

=====
Package           Architecture Version           Repository        Size
=====
Installing:
dovecot            x86_64         1:2.3.7.2-1.fc29 updates          4.6 M
postfix            x86_64         2:3.3.3-1.fc29  updates          1.4 M
Installing dependencies:
libsodium          x86_64         1.0.18-1.fc29  updates          155 k
=====

Transaction Summary
=====
Install 3 Packages

Total download size: 6.1 M
Installed size: 22 M
Downloading Packages:
(1/3): libsodium-1.0.18-1.fc29.x86_64.rpm 3.1 kB/s | 155 kB 00:49
[MIRROR] dovecot-2.3.7.2-1.fc29.x86_64.rpm: Curl error (28): Timeout was reached
for https://fedora-archive.ip-connect.info/fedora/linux/updates/29/Everything/x
86_64/Packages/d/dovecot-2.3.7.2-1.fc29.x86_64.rpm [Operation too slow. Less tha
n 1000 bytes/sec transferred the last 30 seconds]
[MIRROR] dovecot-2.3.7.2-1.fc29.x86_64.rpm: Curl error (28): Timeout was reached
for http://fedora-archive.ip-connect.info/fedora/linux/updates/29/Everything/x8
6_64/Packages/d/dovecot-2.3.7.2-1.fc29.x86_64.rpm [Operation too slow. Less than
1000 bytes/sec transferred the last 30 seconds]
[MIRROR] postfix-3.3.3-1.fc29.x86_64.rpm: Curl error (28): Timeout was reached f
```

```
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing      : 1/1
Installing     : libsodium-1.0.18-1.fc29.x86_64 1/3
Running scriptlet: dovecot-1:2.3.7.2-1.fc29.x86_64 2/3
Installing     : dovecot-1:2.3.7.2-1.fc29.x86_64 2/3
Running scriptlet: dovecot-1:2.3.7.2-1.fc29.x86_64 2/3
Running scriptlet: postfix-2:3.3.3-1.fc29.x86_64 3/3
Installing     : postfix-2:3.3.3-1.fc29.x86_64 3/3
Running scriptlet: postfix-2:3.3.3-1.fc29.x86_64 3/3
Running scriptlet: dovecot-1:2.3.7.2-1.fc29.x86_64 3/3
Running scriptlet: postfix-2:3.3.3-1.fc29.x86_64 3/3
Verifying      : dovecot-1:2.3.7.2-1.fc29.x86_64 1/3
Verifying      : libsodium-1.0.18-1.fc29.x86_64 2/3
Verifying      : postfix-2:3.3.3-1.fc29.x86_64 3/3

Installed:
dovecot-1:2.3.7.2-1.fc29.x86_64 postfix-2:3.3.3-1.fc29.x86_64
libsodium-1.0.18-1.fc29.x86_64

Complete!
[student@localhost ~]$
```


Project Realization

1. Installation

- Spamassassin was installed to enhance email security by filtering and detecting spam messages.

```
Complete!
[student@localhost ~]$ sudo dnf install spamassassin spamass-milter -y
[sudo] password for student:
Last metadata expiration check: 1:23:34 ago on Tue 28 Apr 2026 10:36:15 AM +03.
Dependencies resolved.
```

Package	Arch	Version	Repository	Size
Installing:				
spamass-milter	x86_64	0.4.0-14.fc29	fedora	61 k
spamassassin	x86_64	3.4.2-2.fc29	fedora	1.2 M
Installing dependencies:				
perl-DB_File	x86_64	1.852-1.fc29	updates	79 k
perl-HTTP-Date	noarch	6.04-1.fc29	updates	23 k
perl-IO-Zlib	noarch	1:1.10-435.fc29	updates	27 k
perl-LWP-MediaTypes	noarch	6.04-1.fc29	updates	33 k
perl-Math-BigInt	noarch	1:1.9998.13-4.fc29	updates	193 k
perl-Math-Complex	noarch	1.59-435.fc29	updates	55 k
perl-Net-DNS	noarch	1.19-1.fc29	updates	361 k
perl-Net-HTTP	noarch	6.19-1.fc29	updates	40 k
perl-Socket6	x86_64	0.29-1.fc29	updates	30 k
perl-Time-HiRes	x86_64	1.9760-1.fc29	updates	57 k
perl-Algorithm-Diff	noarch	1.1903-11.fc29	fedora	48 k

Project Realization

2. Configuration

Postfix configuration was verified and tested to ensure correct settings and proper operation.

```
student@localhost:~  
File Edit View Search Terminal Help  
[student@localhost ~]$ sudo systemctl restart postfix  
[sudo] password for student:  
Job for postfix.service failed because the control process exited with error code.  
See "systemctl status postfix.service" and "journalctl -xe" for details.  
[student@localhost ~]$ salam.com  
bash: salam.com: command not found...  
[student@localhost ~]$ sudo postfix check  
postfix: warning: /etc/postfix/main.cf, line 743: overriding earlier entry: inet_interfaces=localhost  
postfix: warning: /etc/postfix/main.cf, line 744: overriding earlier entry: mydestination=$myhostname, localhost.$mydomain, localhost  
postfix: fatal: /etc/postfix/main.cf, line 746: missing '=' after attribute name: "sudo systemctl restart postfix"
```

Project Realization

2. Configuration

- Configuring Postfix by editing the main.cf file and verifying the configuration using postfix check to detect and fix errors.

```
[student@localhost ~]$ sudo nano /etc/postfix/main.cf
[student@localhost ~]$ sudo postfix check
postfix: warning: /etc/postfix/main.cf, line 742: overriding earlier entry: inet_interfaces=localhost
postfix: warning: /etc/postfix/main.cf, line 743: overriding earlier entry: mydestination=$myhostname, localhost.$mydomain, localhost
/usr/libexec/postfix/postfix-script: line 74: cd: too many arguments
postfix/postlog: warning: /etc/postfix/main.cf, line 742: overriding earlier entry: inet_interfaces=localhost
postfix/postlog: warning: /etc/postfix/main.cf, line 743: overriding earlier entry: mydestination=$myhostname, localhost.$mydomain, localhost
postfix/postfix-script: fatal: no Postfix shared-library directory /usr/lib64/postfixmyhostname = mail.salam.com!
[student@localhost ~]$ sudo systemctl restart postfix
Job for postfix.service failed because the control process exited with error code.
See "systemctl status postfix.service" and "journalctl -xe" for details.
[student@localhost ~]$
```


Project Realization

2. Configuration

Using commands like postfix check and systemctl restart postfix to identify configuration errors and verify the service status.

```
student@localhost:~
File Edit View Search Terminal Help
[student@localhost ~]$ sudo systemctl restart postfix
[sudo] password for student:
Job for postfix.service failed because the control process exited with error code.
See "systemctl status postfix.service" and "journalctl -xe" for details.
[student@localhost ~]$ salam.com
bash: salam.com: command not found...
[student@localhost ~]$ sudo postfix check
postfix: warning: /etc/postfix/main.cf, line 743: overriding earlier entry: inet_interfaces=localhost
postfix: warning: /etc/postfix/main.cf, line 744: overriding earlier entry: mydestination=$myhostname, localhost.$mydomain, localhost
postfix: fatal: /etc/postfix/main.cf, line 746: missing '=' after attribute name: "sudo systemctl restart postfix"
[student@localhost ~]$ sudo nano /etc/postfix/main.cf
[student@localhost ~]$ sudo postfix check
postfix: warning: /etc/postfix/main.cf, line 742: overriding earlier entry: inet_interfaces=localhost
postfix: warning: /etc/postfix/main.cf, line 743: overriding earlier entry: mydestination=$myhostname, localhost.$mydomain, localhost
/usr/libexec/postfix/postfix-script: line 74: cd: too many arguments
postfix/postlog: warning: /etc/postfix/main.cf, line 742: overriding earlier entry: inet_interfaces=localhost
postfix/postlog: warning: /etc/postfix/main.cf, line 743: overriding earlier entry: mydestination=$myhostname, localhost.$mydomain, localhost
postfix/postfix-script: fatal: no Postfix shared-library directory /usr/lib64/postfix
stfmyhostname = mail.salam.com!
[student@localhost ~]$
```

Project Realization

2.Configuration

- Restarting the Postfix service and verifying its status to ensure it is running successfully.

```
student@localhost:~
File Edit View Search Terminal Help
[student@localhost ~]$ sudo systemctl restart postfix
[sudo] password for student:
[student@localhost ~]$ sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2026-04-28 12:54:47 +03; 21s ago
     Process: 19802 ExecStop=/usr/sbin/postfix stop (code=exited, status=0/SUCCESS)
     Process: 19818 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)
     Process: 19817 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)
     Process: 19815 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)
   Main PID: 19885 (master)
      Tasks: 3 (limit: 4915)
     Memory: 7.4M
    CGroup: /system.slice/postfix.service
            └─19885 /usr/libexec/postfix/master -w
              └─19886 pickup -l -t unix -u
                └─19887 qmgr -l -t unix -u

Apr 28 12:54:47 localhost.localdomain postfix/postlog[19883]: warning: /etc/postfix/main.cf, line 744: overriding earlier entry: mydestination=
Apr 28 12:54:47 localhost.localdomain postfix/postfix-script[19883]: starting the Postfix mail system
Apr 28 12:54:47 localhost.localdomain postfix/master[19885]: warning: /etc/postfix/main.cf, line 743: overriding earlier entry: inet_interfaces=
Apr 28 12:54:47 localhost.localdomain postfix/master[19885]: warning: /etc/postfix/main.cf, line 744: overriding earlier entry: mydestination=
Apr 28 12:54:47 localhost.localdomain postfix/master[19885]: daemon started -- version 3.3.3, configuration /etc/postfix
Apr 28 12:54:47 localhost.localdomain systemd[1]: Started Postfix Mail Transport Agent.
Apr 28 12:54:47 localhost.localdomain postfix/qmgr[19887]: warning: /etc/postfix/main.cf, line 743: overriding earlier entry: inet_interfaces=
Apr 28 12:54:47 localhost.localdomain postfix/qmgr[19887]: warning: /etc/postfix/main.cf, line 744: overriding earlier entry: mydestination=$m
Apr 28 12:54:47 localhost.localdomain postfix/qmgr[19887]: warning: /etc/postfix/main.cf, line 744: overriding earlier entry: mydestination=$m
lines 1-23...skipping...
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2026-04-28 12:54:47 +03; 21s ago
     Process: 19802 ExecStop=/usr/sbin/postfix stop (code=exited, status=0/SUCCESS)
     Process: 19818 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)
     Process: 19817 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)
```


Project Realization

2. Configuration

Creating a mail user account and setting up the Maildir structure for email storage.

```
Process: 20109 ExecStop=/usr/bin/doveadm stop (code=exited, status=0/SUCCESS)
Process: 20113 ExecStartPre=/usr/libexec/dovecot/prestartscript (code=exited, status=0)
Main PID: 20119 (dovecot)
Tasks: 4 (limit: 4915)
Memory: 4.5M
CGroup: /system.slice/dovecot.service
├─20119 /usr/sbin/dovecot -F
├─20126 dovecot/anvil
├─20127 dovecot/log
└─20128 dovecot/config

Apr 28 13:34:37 localhost.localdomain systemd[1]: Starting Dovecot IMAP/POP3 email serv
Apr 28 13:34:37 localhost.localdomain systemd[1]: Started Dovecot IMAP/POP3 email serve
Apr 28 13:34:37 localhost.localdomain dovecot[20119]: master: Dovecot v2.3.7.2 (3c910f6
[student@localhost ~]$ sudo adduser salam
[student@localhost ~]$ sudo passwd salam
Changing password for user salam.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[student@localhost ~]$ su - salam
Password:
[salam@localhost ~]$ maildirmake.dovecot ~/Maildir
bash: maildirmake.dovecot: command not found...
[salam@localhost ~]$ mkdir -p ~/Maildir
[salam@localhost ~]$ mkdir -p ~/Maildir/{new,cur,tmp}
[salam@localhost ~]$ exit
logout
```


Project Realization

3. Security:

Configuring Postfix settings including basic parameters and enabling TLS security for secure email transmission.

```

student@localhost:~
GNU nano 3.0 /etc/postfix/main.cf Modified
#
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt
# Use TLS if this is supported by the remote SMTP server, otherwise use
# plaintext (opportunistic TLS outbound).
#
smtp_tls_security_level = may
meta_directory = /etc/postfix
shlib_directory = /usr/lib64/postfix myhostname = mail.salam.com
mydomain = salam.com
myorigin = $mydomain
inet_interfaces = all
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
home_mailbox = Maildir/

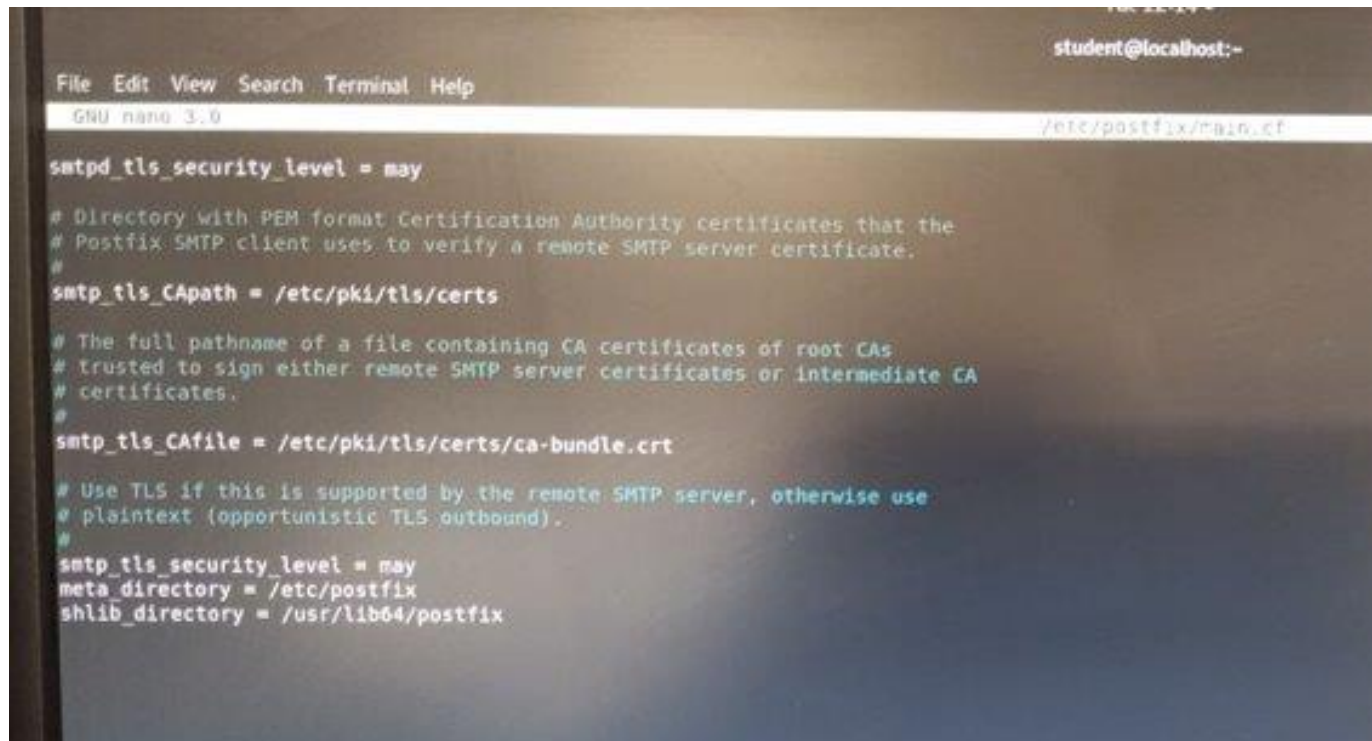
I

^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text   ^J Justify    ^C Cur Pos   M-U Undo     M-A Mark Text M-I To Bracket
^X Exit      ^R Read File  ^\ Replace   ^U Uncut Text ^T To Spell   ^_ Go To Line M-E Redo     M-C Copy Text ^0 Where Was
  
```

Project Realization

3. Security:

Enabling TLS security in Postfix configuration to ensure encrypted and secure email transmission.



```
student@localhost:~  
File Edit View Search Terminal Help  
GNU nano 3.0 /etc/postfix/main.cf  
smtpd_tls_security_level = may  
# Directory with PEM format Certification Authority certificates that the  
# Postfix SMTP client uses to verify a remote SMTP server certificate.  
#  
smtpd_tls_CApath = /etc/pki/tls/certs  
# The full pathname of a file containing CA certificates of root CAs  
# trusted to sign either remote SMTP server certificates or intermediate CA  
# certificates.  
#  
smtpd_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt  
# Use TLS if this is supported by the remote SMTP server, otherwise use  
# plaintext (opportunistic TLS outbound).  
#  
smtpd_tls_security_level = may  
meta_directory = /etc/postfix  
shlib_directory = /usr/lib64/postfix
```

Project Realization

1. Installation

Installing Dovecot mail server package and verifying that it is successfully installed

```
[salam@localhost ~]$ sudo dnf install dovecot -y
[sudo] password for salam:
salam is not in the sudoers file. This incident will be reported.
[salam@localhost ~]$ exit
logout
[student@localhost ~]$ sudo dnf install dovecot -y
Last metadata expiration check: 0:26:23 ago on Tue 28 Apr 2026 01:43:56 PM +03.
Package dovecot-1:2.3.7.2-1.fc29.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[student@localhost ~]$ sudo systemctl restart devocot
Failed to restart devocot.service: Unit devocot.service not found.
[student@localhost ~]$
```


Project Realization

2.configuration

Configuring Dovecot and starting the service, then verifying that it is running successfully

```
student@localhost:~  
File Edit View Search Terminal Help  
[student@localhost ~]$ q  
bash: q: command not found...  
[student@localhost ~]$ sudo nano /etc/dovecot/dovecot.conf  
[sudo] password for student:  
[student@localhost ~]$ sudo nano /etc/dovecot/conf.d/10-mail.conf  
[student@localhost ~]$ sudo systemctl restart dovecot  
[student@localhost ~]$ sudo systemctl status dovecot  
● dovecot.service - Dovecot IMAP/POP3 email server  
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; vendor preset: disabled)  
   Active: active (running) since Tue 2026-04-28 13:34:37 +03; 21s ago  
     Docs: man:dovecot(1)  
           http://wiki2.dovecot.org/  
   Process: 20109 ExecStop=/usr/bin/doveadm stop (code=exited, status=0/SUCCESS)  
   Process: 20113 ExecStartPre=/usr/libexec/dovecot/prestartscript (code=exited, status=0/SUCCESS)  
 Main PID: 20119 (dovecot)  
    Tasks: 4 (limit: 4915)  
   Memory: 4.5M  
    CGroup: /system.slice/dovecot.service  
            └─20119 /usr/sbin/dovecot -F  
              └─20126 dovecot/anvil  
                └─20127 dovecot/log  
                  └─20128 dovecot/config  
  
Apr 28 13:34:37 localhost.localdomain systemd[1]: Starting Dovecot IMAP/POP3 email server...  
Apr 28 13:34:37 localhost.localdomain systemd[1]: Started Dovecot IMAP/POP3 email server.  
Apr 28 13:34:37 localhost.localdomain dovecot[20119]: master: Dovecot v2.3.7.2 (3c910f64b) starting up for imap, pop3, lmtp  
[student@localhost ~]$
```

Project Realization

4.Test

Verifying the mail server by sending a test email

```
salam@localhost:~  
  
File Edit View Search Terminal Help  
--E1E641A104D.1777374587@mail.salam.com  
Content-Description: Undelivered Message  
Content-Type: message/rfc822  
Content-Transfer-Encoding: 8bit  
  
Return-Path: <salam@salam.com>  
Received: by mail.salam.com (Postfix, from userid 11125)  
        id E1E641A104D; Tue, 28 Apr 2026 14:09:46 +0300 (+03)  
To: root@salam.com  
From: salam@salam.com  
Auto-Submitted: auto-generated  
Subject: *** SECURITY information for localhost.localdomain ***  
Message-Id: <20260428110946.E1E641A104D@mail.salam.com>  
Date: Tue, 28 Apr 2026 14:09:46 +0300 (+03)  
  
localhost.localdomain : Apr 28 14:09:46 : salam : user NOT in sudoers ; TTY=pts/  
0 ; PWD=/home/salam ; USER=root ; COMMAND=/usr/bin/dnf install dovecot -y  
  
--E1E641A104D.1777374587@mail.salam.com--  
Return-Path: <student@salam.com>  
X-Original-To: salam@localhost  
Delivered-To: salam@localhost  
Received: by mail.salam.com (Postfix, from userid 1000)  
        id 9C4341A103B; Tue, 28 Apr 2026 14:14:57 +0300 (+03)  
Date: Tue, 28 Apr 2026 14:14:57 +0300  
To: salam@localhost  
Subject: Hello  
User-Agent: Heirloom mailx 12.5 7/5/10  
MIME-Version: 1.0  
Content-Type: text/plain; charset=us-ascii  
Content-Transfer-Encoding: 7bit  
Message-Id: <20260428111457.9C4341A103B@mail.salam.com>  
From: student <student@salam.com>  
  
Test mail  
[salam@localhost ~]$
```

References:

<https://dev.to/khripkooleg/lets-build-a-local-mail-server-from-scratch-with-postfix-and-dovecot-3gje>

<https://cubepath.com/docs/email-server/postfix-dovecot-complete-configuration>

<https://medium.com/@vempatisriram6/building-your-own-local-mail-server-using-postfix-dovecot-e6ad56c76be7>